

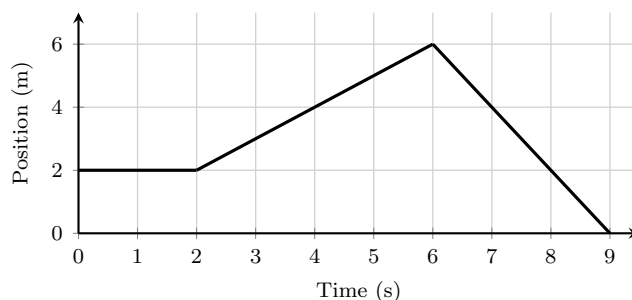
Instructions: Answer all questions. Write in complete sentences unless told otherwise, and show your work for calculations.

Position–Time Graphs

A position–time graph shows **position (m)** on the y -axis and **time (s)** on the x -axis.

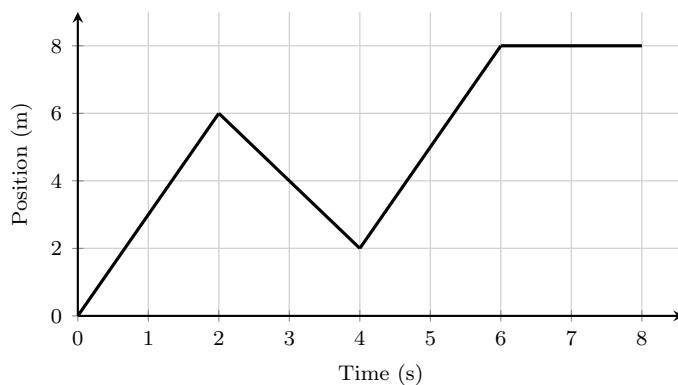
- The **slope** of the line is the **velocity**: $v = \frac{\Delta x}{t}$. A steeper slope means a greater speed.
- A **flat** line (slope = 0) means the object is **at rest**.
- The object **changes direction** when the sign of the slope changes (positive $\leftrightarrow$ negative).
- **Distance** is the total path length (always positive); **displacement** is $\Delta x = x - x_0$ (has a sign).
- $s_{avg} = \frac{d_{total}}{t_{total}}$; $v_{avg} = \frac{\Delta x_{total}}{t_{total}}$.

1. The position–time graph below shows the motion of an object along a straight track.



- For what time interval is the object at rest?
- At what time interval does the object have the greatest speed?
- At what time does the object turn around?
- Calculate the object's total distance traveled from $t = 0$ s to $t = 9$ s.
- Calculate the object's total displacement from $t = 0$ s to $t = 9$ s.

2. The position–time graph below shows the motion of a toy car along a straight track. The graph is made of four straight segments.



- (a) Calculate the car's velocity in each time interval: 0–2 s, 2–4 s, 4–6 s, and 6–8 s. Show your use of $v = \frac{x-x_0}{t}$ and include the sign and the units.
- (b) Calculate the car's total distance traveled for the whole trip ($t = 0$ to $t = 8$ s).
- (c) Calculate the car's total displacement for the whole trip.
- (d) Calculate the car's average speed for the whole trip.
- (e) Calculate the car's average velocity for the whole trip.

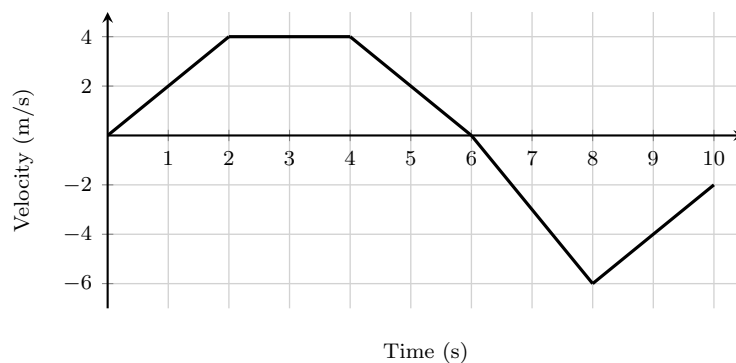
Velocity–Time Graphs

A velocity–time graph shows **velocity (m/s)** on the y -axis and **time (s)** on the x -axis. The y -value at a given time is the velocity at that time.

Reminders

- The **slope** of the line is the **acceleration**: $a = \frac{\Delta v}{\Delta t}$.
- A **steeper** slope means a **greater acceleration**.
- A **flat** line (slope = 0) means **constant velocity** (acceleration = 0).
- If the line moves **away from** the time axis, the speed is increasing: the object is **speeding up**.
- If the line moves **toward** the time axis, the speed is decreasing: the object is **slowing down**.
- When the line **crosses** the time axis ($v = 0$), the object **changes direction**.

3. The velocity–time graph below shows the motion of an object along a straight line. The graph is made of five straight segments.



- During which time interval(s) is the object speeding up?
- During which time interval(s) is the object slowing down?
- During which time interval(s) does the object have a constant velocity?
- At what time(s) does the object change direction?
- Calculate the acceleration of the object from $t = 0$ s to $t = 2$ s.
- Which time interval has the greatest acceleration? Explain how you can tell from the graph.